



**BPM 1  
CPR**

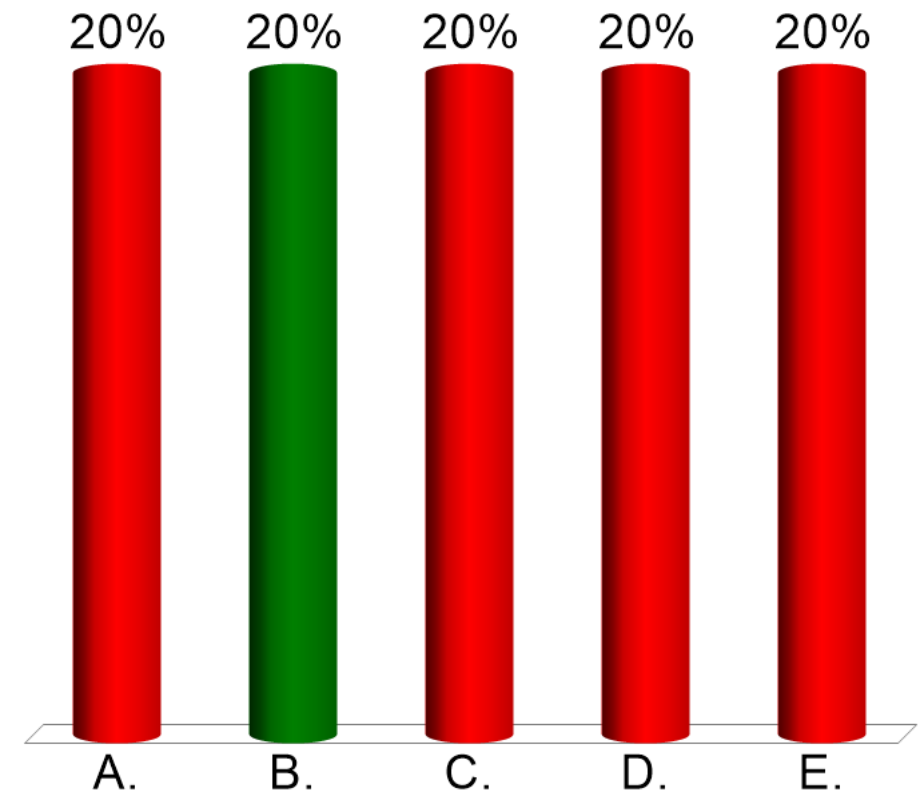
**IMCQ 2  
PHYSIOLOGY  
Fall 2025**

1. A 70-year-old woman is brought to the emergency department because of a sudden onset of shortness of breath for the past hour. She has hypertension and stable angina. Physical examination shows bibasilar crackles and an S3 gallop. She is treated with high-dose intravenous nitroglycerin, and her symptoms improve. Which of the following hemodynamic parameters best reflects a reduction in left ventricular afterload associated with this therapy?

- A. Left ventricular end-diastolic pressure
- B. Mean systemic arterial pressure
- C. Pulmonary capillary wedge pressure
- D. Right atrial pressure
- E. Venous return to the right atrium

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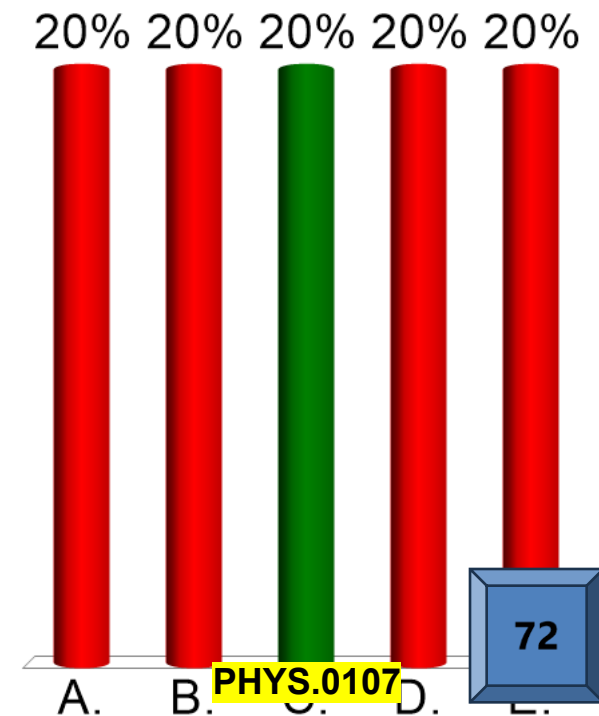
Define afterload and explain how arterial pressure increases or decreases afterload.

2. A 52-year-old man comes to the emergency room because of progressive shortness of breath and fatigue for the past 3 hours. Physical examination shows a holosystolic murmur best heard at the cardiac apex radiating to the axilla and bibasilar crackles. Echocardiographic and angiographic data show regurgitant flow from the left ventricle into the left atrium during systole. Which of the following changes is most likely observed on this patient's Wigger's diagram during ventricular systole?

- A. Elevated left ventricular end-diastolic pressure due to impaired relaxation
- B. Equalization of left atrial and left ventricular pressures during diastole
- C. Prominent V-wave due to increased left atrial filling during systole
- D. Rapid decline in aortic pressure during diastole
- E. Reduced left atrial pressure during ventricular contraction

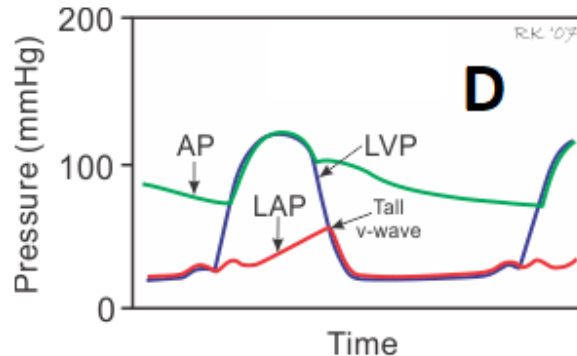
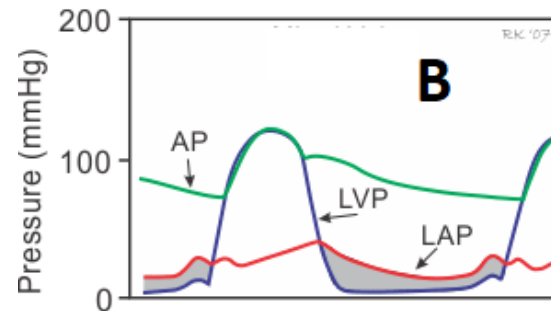
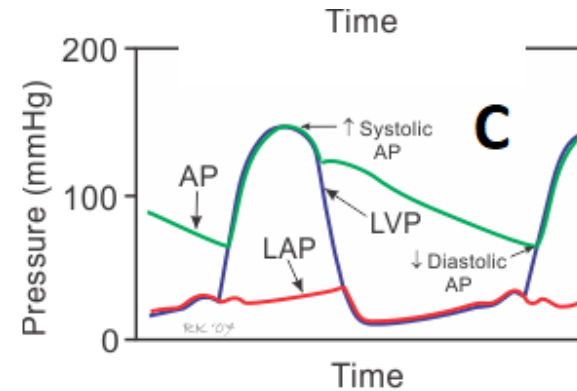
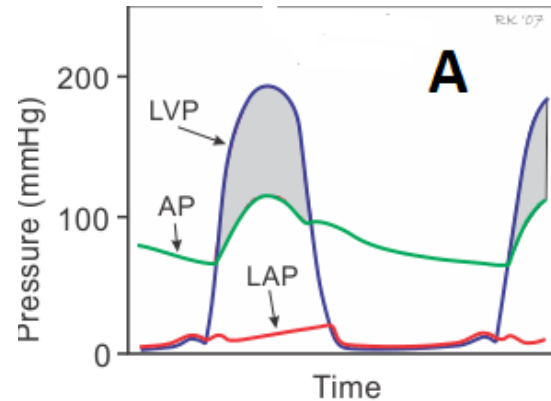
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3. A 37-year-old woman comes to the family clinic for a routine health maintenance examination. She has no complaints apart from occasional shortness of breath. Cardiac examination shows a "snap"- like sound after S2, followed by a rumbling murmur that is heard best at the 5th Intercostal space, mid-clavicular line. Which of the following cardiac catheterization measurements best represents this patient?

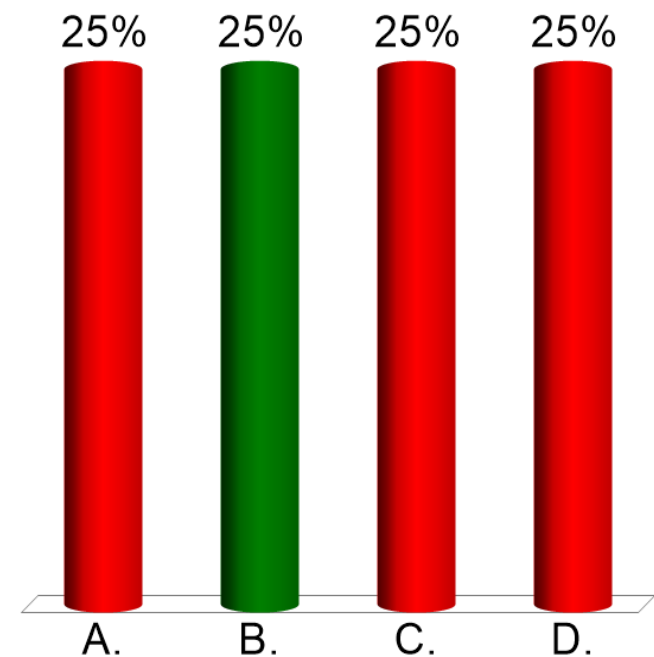
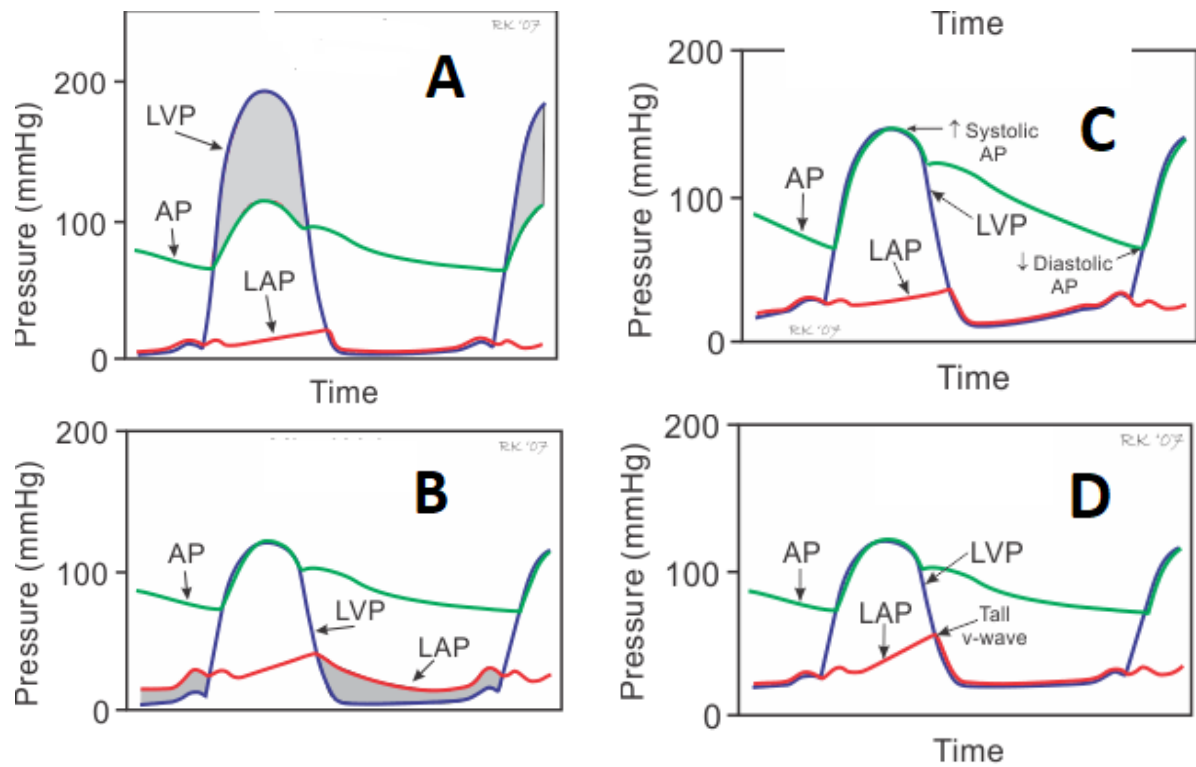
- A. A
- B. B
- C. C
- D. D



PHYS.0107

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- B. B
- C. C
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PHYS.0107



4. A 25-year-old man comes to the physician because of a 1-month history of worsening exertional dyspnea and intermittent leg edema. Physical examination shows a II/IV early diastolic murmur loudest at the second left intercostal space, parasternally. An x-ray of the chest shows a slight increase in heart size. The graph shown demonstrates the hemodynamics found during cardiac catheterization of the right ventricle and pulmonary artery. Which of the following findings is most likely in this patient?

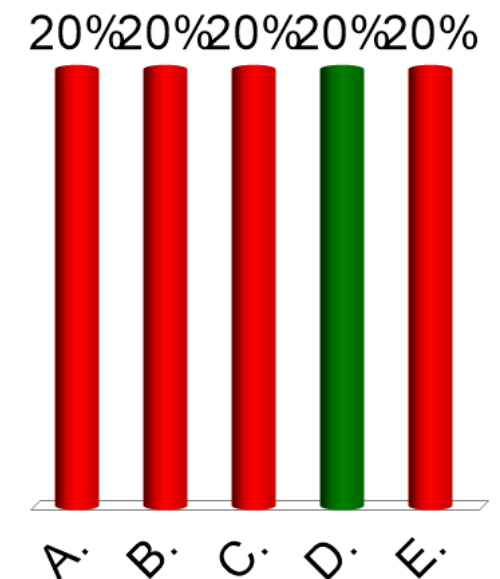
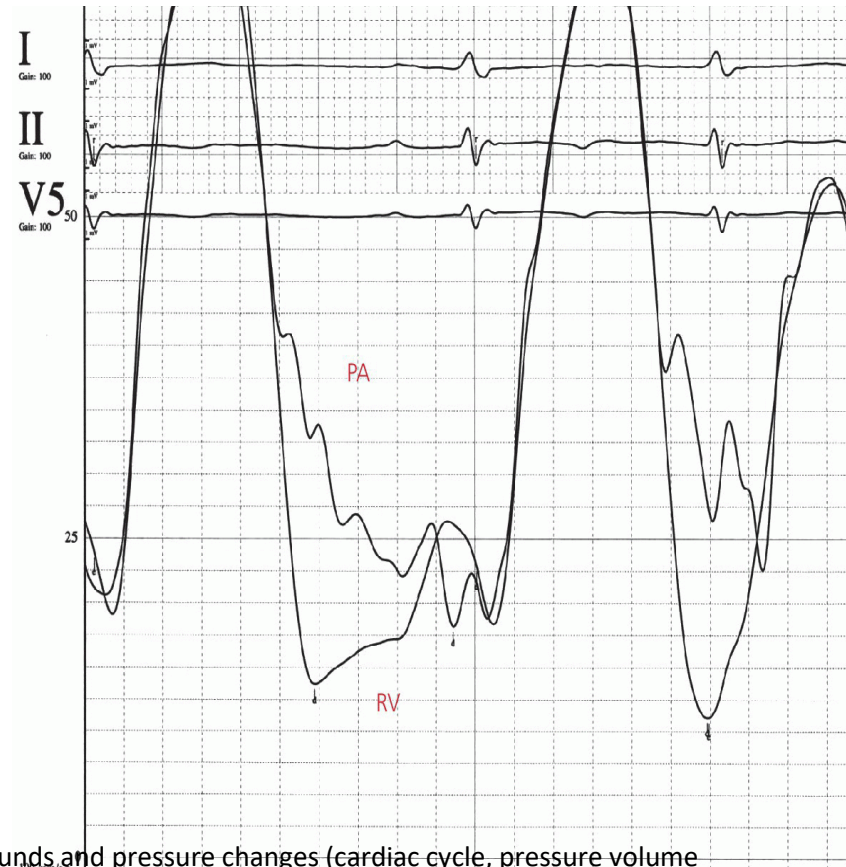
- A. Regurgitation of blood from the ventricle into the atrium occurs during systole.
- B. Right atrial pressure exceeds pulmonary arterial pressure during diastole.
- C. Retrograde flow of blood from the aorta into the left ventricle occurs during relaxation.
- D. Pulmonary arterial diastolic pressure declines during ventricular relaxation.
- E. Right ventricular pressure surpasses pulmonary arterial pressure during ejection.





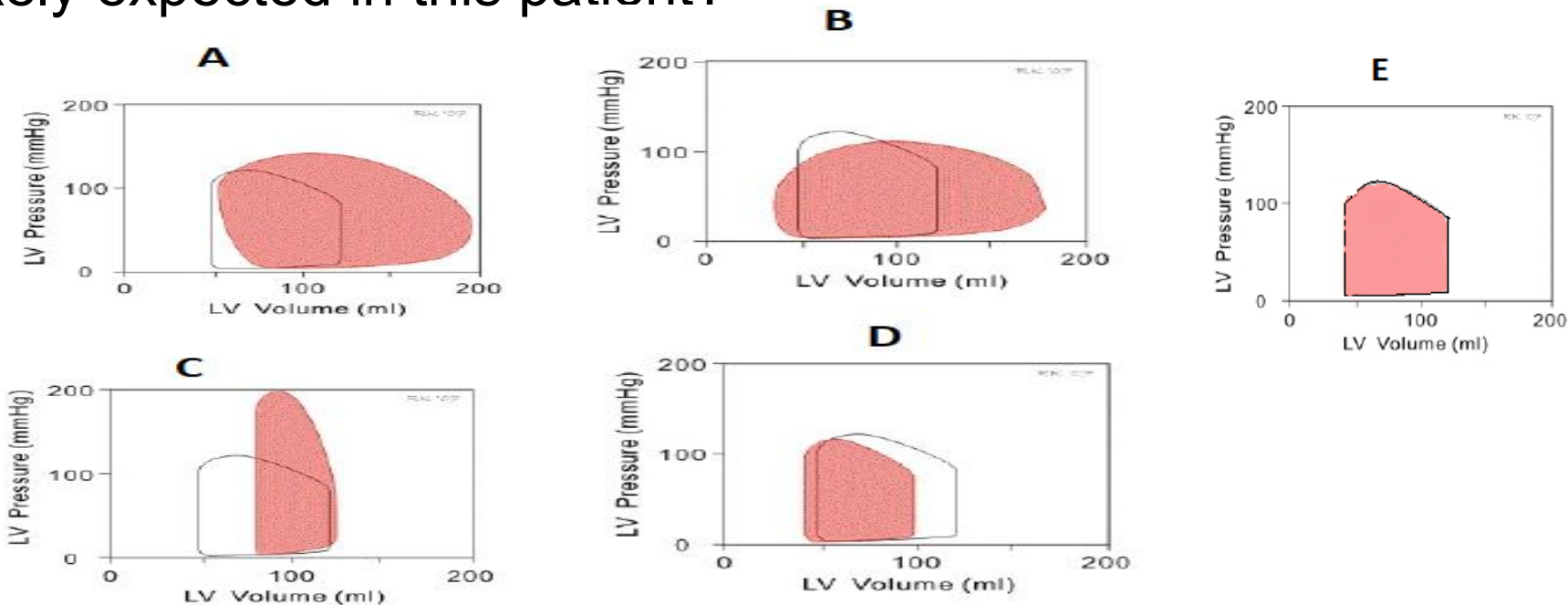
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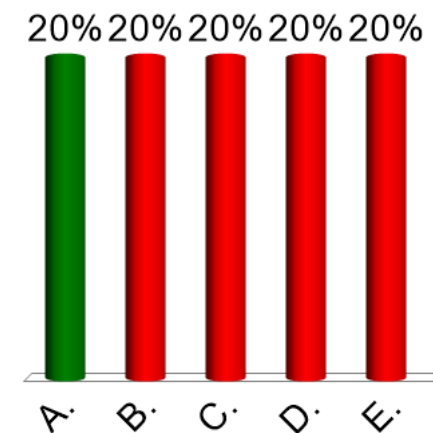
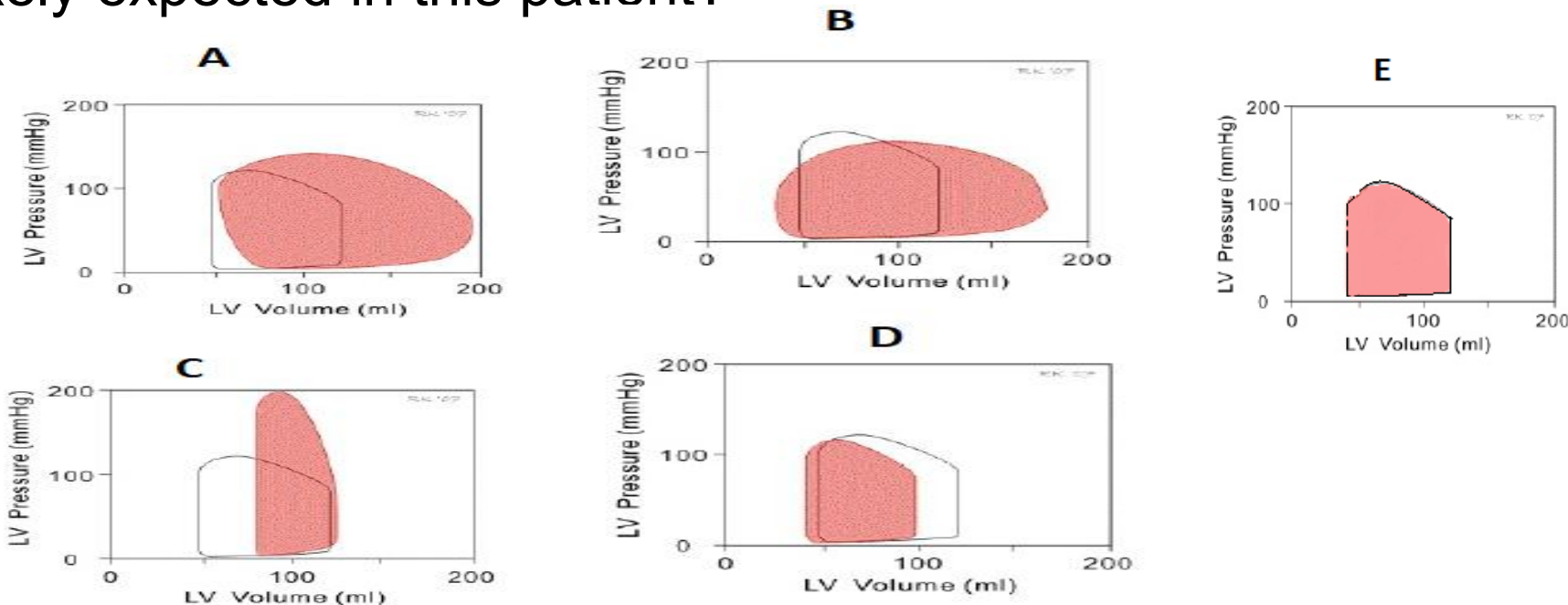
5. A 43-year-old woman comes to the physician because of a 3-day history of right buttock pain. She is an active IV heroin user and had a history of recent aortic valve endocarditis associated with *Streptococcus milleri* infection, which was treated with six weeks of antibiotics. Temperature is 39.1° C (102.4°F) and blood pressure is 143/45 mm Hg. Cardiac examination shows a III/IV diastolic murmur over the base of the heart. Upon applying gentle pressure to the nail tip, alternate filling and blanching of the capillary bed concomitant with the cardiac cycle is observed. Which of the following pressure-volume loops shown in the image is most likely expected in this patient?

- A.A
- B.B
- C.C
- D.D
- E.E



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- D.D
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Describe the expected auscultation sounds and pressure changes (cardiac cycle, pressure volume loops) that define mitral stenosis, mitral insufficiency, aortic stenosis, and aortic insufficiency.

6. A 68-year-old woman is brought to the emergency department because of 1-hour weakness of her right arm and leg. She has a history of long-standing hypertension and hyperlipidemia. Carotid auscultation shows a right-sided bruit. MRI of the brain demonstrates an ischemic infarct within the middle cerebral artery territory, and angiography shows approximately 70% narrowing of the right internal carotid artery. Assuming mean arterial pressure remains constant, which of the following sets of patterns of physiological changes is most likely to occur distal to the site of obstruction?

|   | Blood flow | Cerebral arterial resistance | Capillary Hydrostatic pressures |
|---|------------|------------------------------|---------------------------------|
| A | increases  | increases                    | increases                       |
| B | increases  | decreases                    | increases                       |
| C | increases  | increases                    | decreases                       |
| D | increases  | decreases                    | decreases                       |
| E | decreases  | increases                    | increases                       |
| F | decreases  | decreases                    | increases                       |
| G | decreases  | increases                    | decreases                       |
| H | decreases  | decreases                    | decreases                       |

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| F | decreases  | decreases                    | increases                       |
| G | decreases  | increases                    | decreases                       |
| H | decreases  | decreases                    | decreases                       |



7. A 56-year-old man is being evaluated for worsening exertional leg pain that improves with rest. Doppler ultrasound of the lower extremities shows a 70% stenosis of the left femoral artery and normal flow through the right leg. Following a supervised exercise program and medical therapy, his symptoms improve. Blood flow and pressures are remeasured in both lower limbs. Which of the following sets of changes is most likely to be found distal to the stenosis in the left leg once a new steady state is achieved?

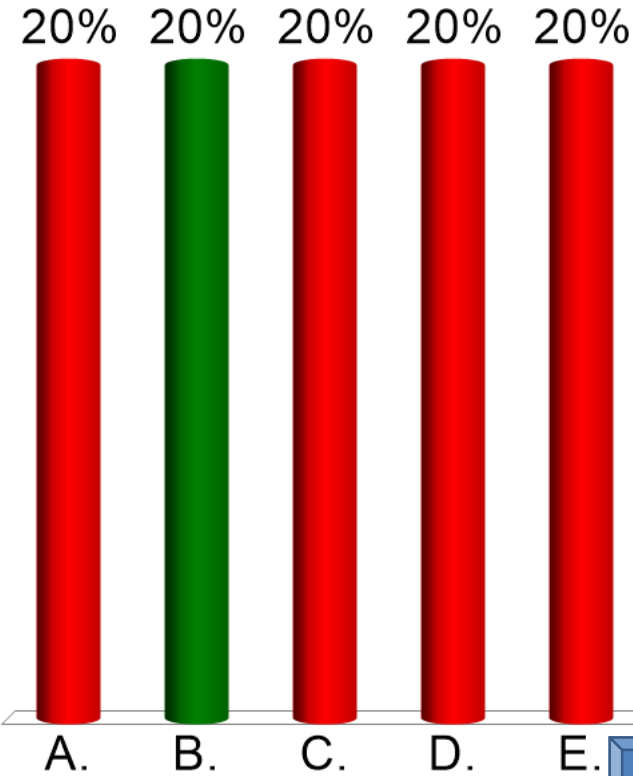
|          | <b>Femoral Artery<br/>Blood Flow</b> | <b>Perfusion<br/>Pressure</b> | <b>Vascular<br/>Conductance</b> |
|----------|--------------------------------------|-------------------------------|---------------------------------|
| <b>A</b> | Increased                            | Increased                     | Increased                       |
| <b>B</b> | Increased                            | Decreased                     | Increased                       |
| <b>C</b> | Decreased                            | Decreased                     | Decreased                       |
| <b>D</b> | Decreased                            | Increased                     | Decreased                       |
| <b>E</b> | Decreased                            | No change                     | Increased                       |



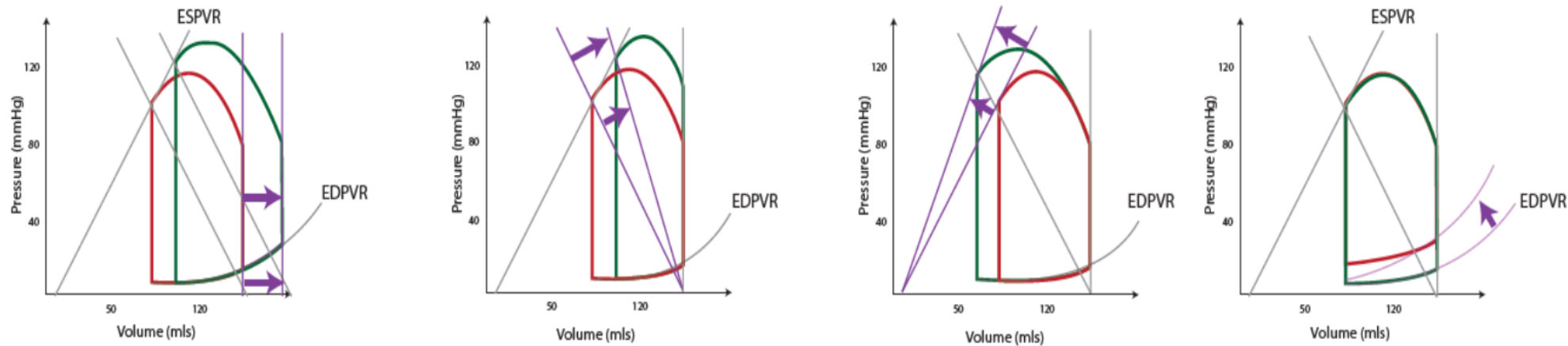
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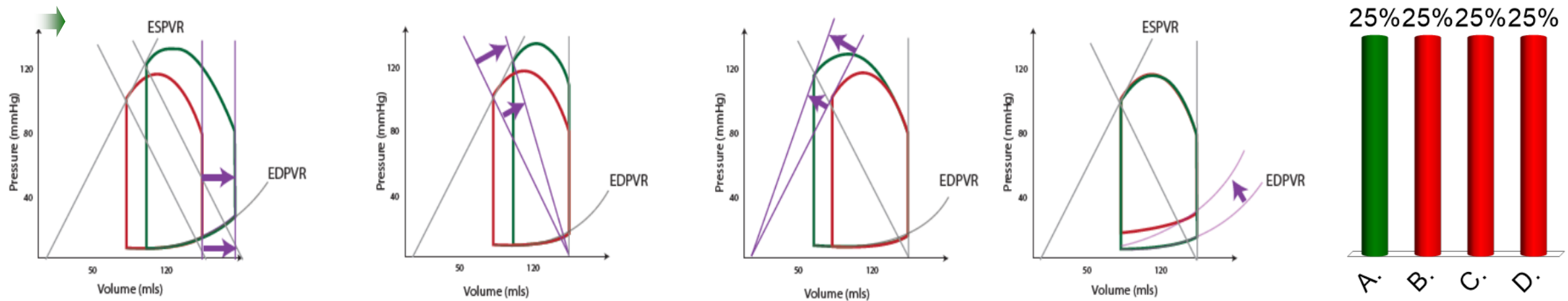
|   | Femoral Artery Blood Flow | Perfusion Pressure | Vascular Conductance |
|---|---------------------------|--------------------|----------------------|
| A | Increased                 | Increased          | Increased            |
| B | Increased                 | Decreased          | Increased            |
| C | Decreased                 | Decreased          | Decreased            |
| D | Decreased                 | Increased          | Decreased            |
| E | Decreased                 | No change          | Increased            |



8. A 48-year-old woman is brought to the emergency department after experiencing profuse vomiting and diarrhea for the past three days due to viral gastroenteritis. She is dehydrated on arrival and receives 3 liters of isotonic saline intravenously over 4 hours. After fluid resuscitation, her blood pressure rises to 125/80 mmHg, and her heart rate decreases to 88 bpm. Repeat examination shows improved capillary refill and jugular venous distension at 30 degrees. She reports feeling better. Which of the following changes in the pressure-volume loop shown in the image below is most likely to be observed after the fluid administration? (red loop before treatment, green loop after treatment)



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9. A 40-year-old woman with a history of repaired atrial septal defect comes to the clinic for a follow-up evaluation. She reports mild fatigue but no chest pain or shortness of breath. As part of her assessment, she undergoes right-heart catheterization, which yields the following data:

Pulmonary artery  $O_2$  content = 12 mL  $O_2$  / 100 mL blood

Pulmonary vein  $O_2$  content = 18 mL  $O_2$  / 100 mL blood

Oxygen consumption = 420 mL  $O_2$  / min

Heart rate = 80 beats/min

Which of the following is the best estimate of this patient's cardiac output?

- A. 3.5 L/min
- B. 4.0 L/min
- C. 5.0 L/min
- D. 6.0 L/min
- E. 7.0 L/min

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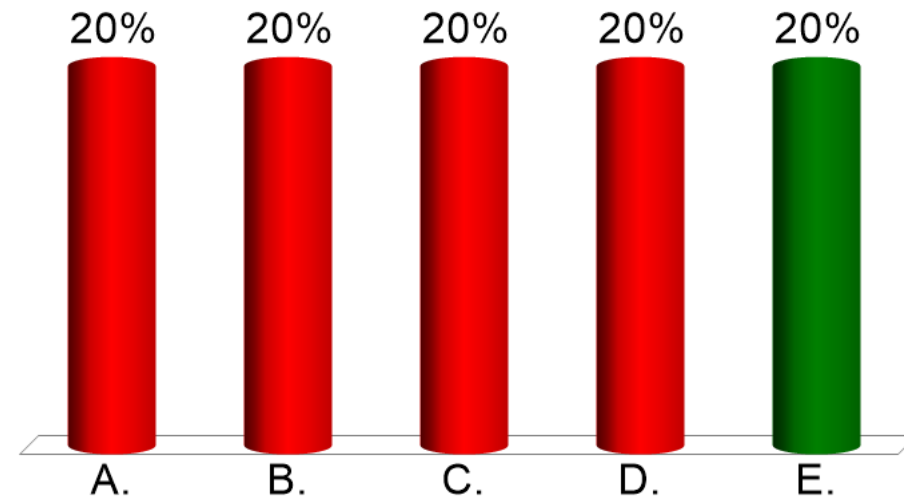
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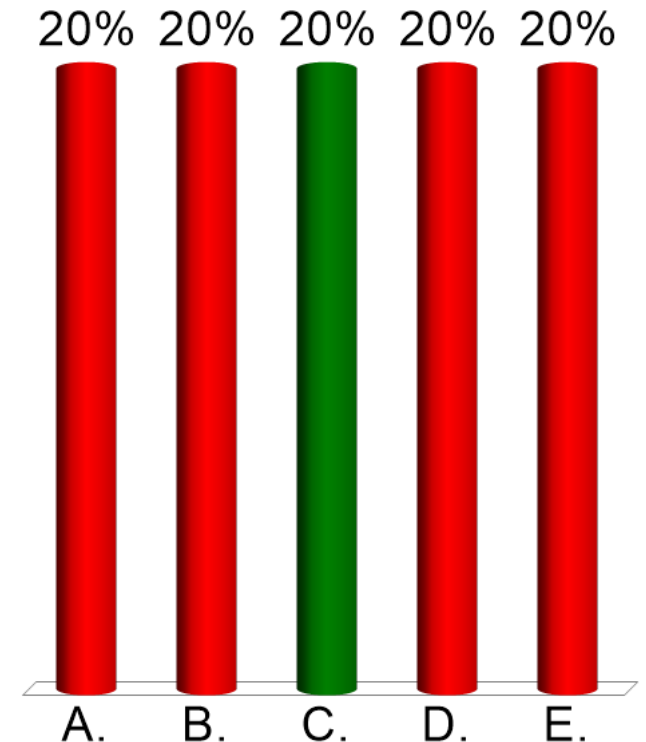
10. A 25-year-old man works outdoors during winter and notices that his fingers become pale and cold after several minutes of exposure. He begins to shiver but otherwise feels well. Physical examination shows cool, dry skin on his hands and forearms. Which of the following hemodynamic changes is most likely to occur in his cutaneous circulation during this response to cold exposure?

- A. Decreased arteriolar resistance in the skin
- B. Increased capillary hydrostatic pressure in the skin
- C. Increased systemic afterload due to sympathetic vasoconstriction
- D. Decreased total peripheral resistance due to  $\beta_2$ -mediated vasodilation
- E. Increased venous compliance leading to blood pooling in the periphery



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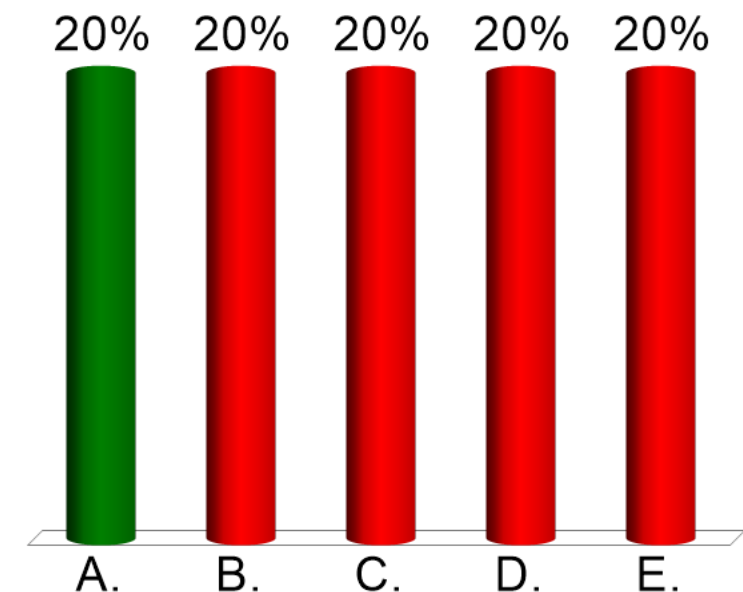
Describe how arterial systolic, diastolic, mean, and pulse pressure are affected by changes in a) stroke volume, b) heart rate, c) arterial compliance, and d) total peripheral resistance.

11. A 45-year-old man comes to the clinic because of a 1-month history of progressive shortness of breath and palpitations. He reports occasional episodes of dizziness when standing quickly. Physical examination show a visible carotid pulsation, a “head bob” synchronous with the heartbeat, and a widened pulse pressure. Cardiac auscultation shows a high-pitched, blowing decrescendo diastolic murmur best heard along the left sternal border when the patient leans forward at end-expiration. Echocardiography shows increased left ventricular end-diastolic volume and retrograde flow during diastole. Which of the following valvular lesions best explains these findings?

- A. Aortic regurgitation
- B. Aortic stenosis
- C. Mitral regurgitation
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PHYS.0120

## Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal (CPR) | IMCQ 1

# CPR IMCQ 2 Anatomy

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**1. A 52-year-old woman comes to the physician because of a 3-month history of fatigue and difficulty breathing when lying flat. She reports some swelling in her legs. She has a history of rheumatoid arthritis, that is controlled by methotrexate. She is 160 cm (5 ft 3 in) tall, weighs 72 kg (159 lb), with a BMI of 28.1. Her temperature is 37°C (98.6°F), pulse is 110/min, respirations are 20/min, and blood pressure is 140/90 mm Hg during expiration and drops to 134/86 during inspiration. Physical examination shows decreased breath sounds at the lung bases, and pitting edema in both legs. Echocardiography shows fluid accumulation around the heart. Which of the following conditions is most likely the cause of her symptoms?**

- A. Cardiac tamponade
- B. Pericardial effusion
- C. Pleural effusion
- D. Mitral stenosis
- E. Pulmonary embolism

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SOM.MK.II.BPM1.3.CPR.2.ANAT.1421 Describe the anatomical location, mechanism and clinical sequences of pericarditis, pericardial friction rub and pericardial effusion. (SG )



**2. A full-term male neonate is brought to the intensive care unit immediately after delivery because of mild rapid breathing and bluish discoloration of the skin noticeable on his lips, hands, and feet. Prenatal ultrasonography showed the absence of the spiral twist in the aortico-pulmonary septum. His pulse is 140/min and respirations are 60/min. Oxygen saturation is 72%. Physical examination shows cyanosis affecting the hands, feet, lips, and tongue. Bedside echocardiography confirms the diagnosis of a cyanotic congenital heart disease with a small patent foramen ovale and a small patent ductus arteriosus. Definitive corrective surgery for the congenital abnormality is planned. Which of the following temporary measures would most likely improve this patient's condition?**

- A. Ablating the sinoatrial node
- B. Administering 100% oxygen
- C. Administering NSAID (indomethacin)
- D. Administering prostaglandin E1 (alprostadil)
- E. Ligating the patent ductus arteriosus
- F. Closing the patent foramen ovale

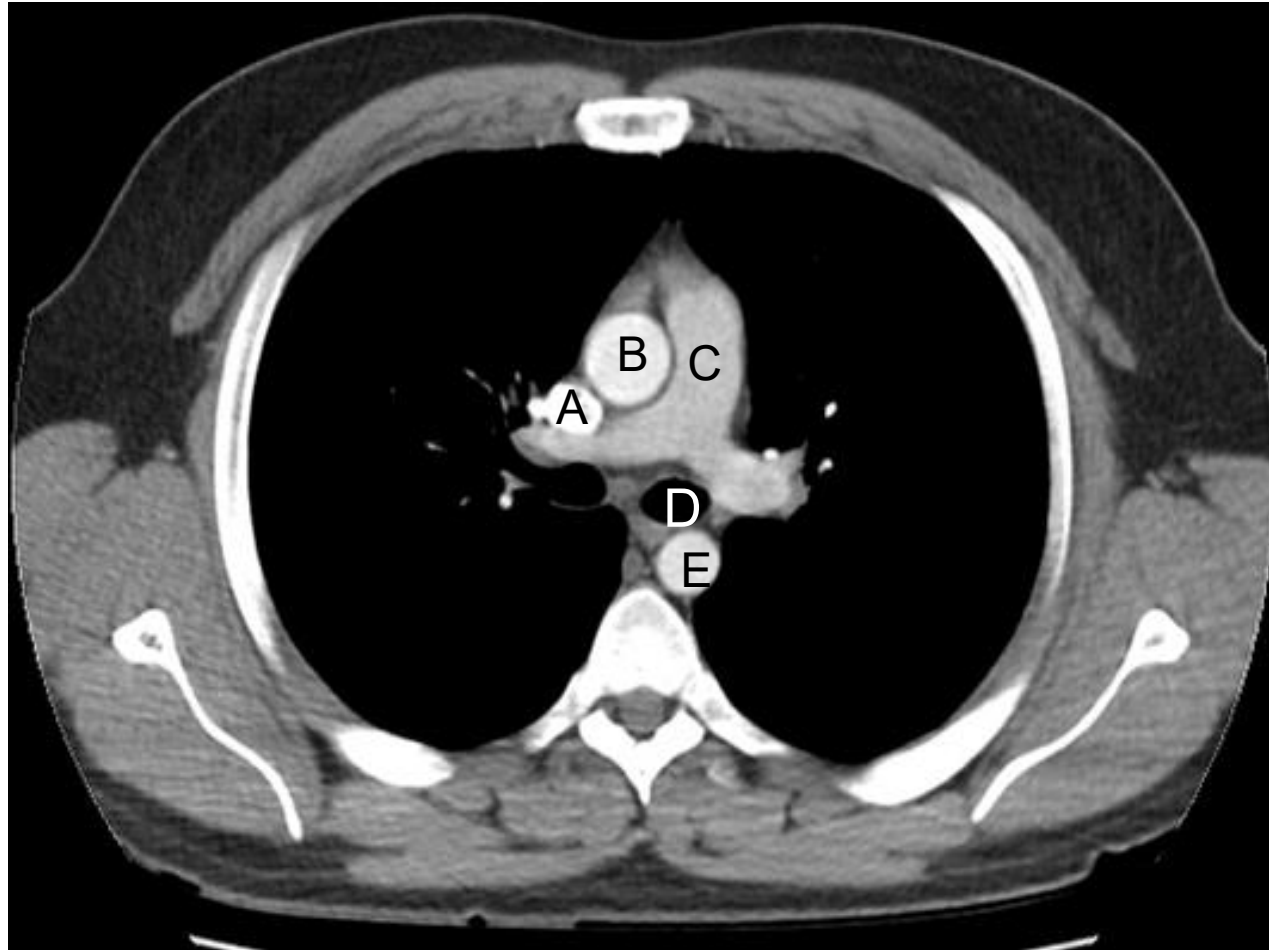
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|                                      |                                                                                                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SOM.MK.II.BPM1.3.C<br>PR.2.ANAT.1420 | Describe the developmental mechanisms underlying ectopia cordis, persistent truncus arteriosus, and transposition of the great arteries, tetralogy of Fallot, total anomalous pulmonary venous return and patent ductus |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**3. A 72-year-old man is brought to the emergency department because of a 2-hour history of intense midback pain, nausea, sweating, and feelings of faintness. Blood pressure measured in the right arm is markedly elevated at 220/105 mm Hg. A transesophageal echocardiography shows the presence of an aortic dissection flap. A CT scan of a normal chest is shown, the affected vessel is the one labelled E. Which of the following is the most likely embryological origin of the affected vessel?**

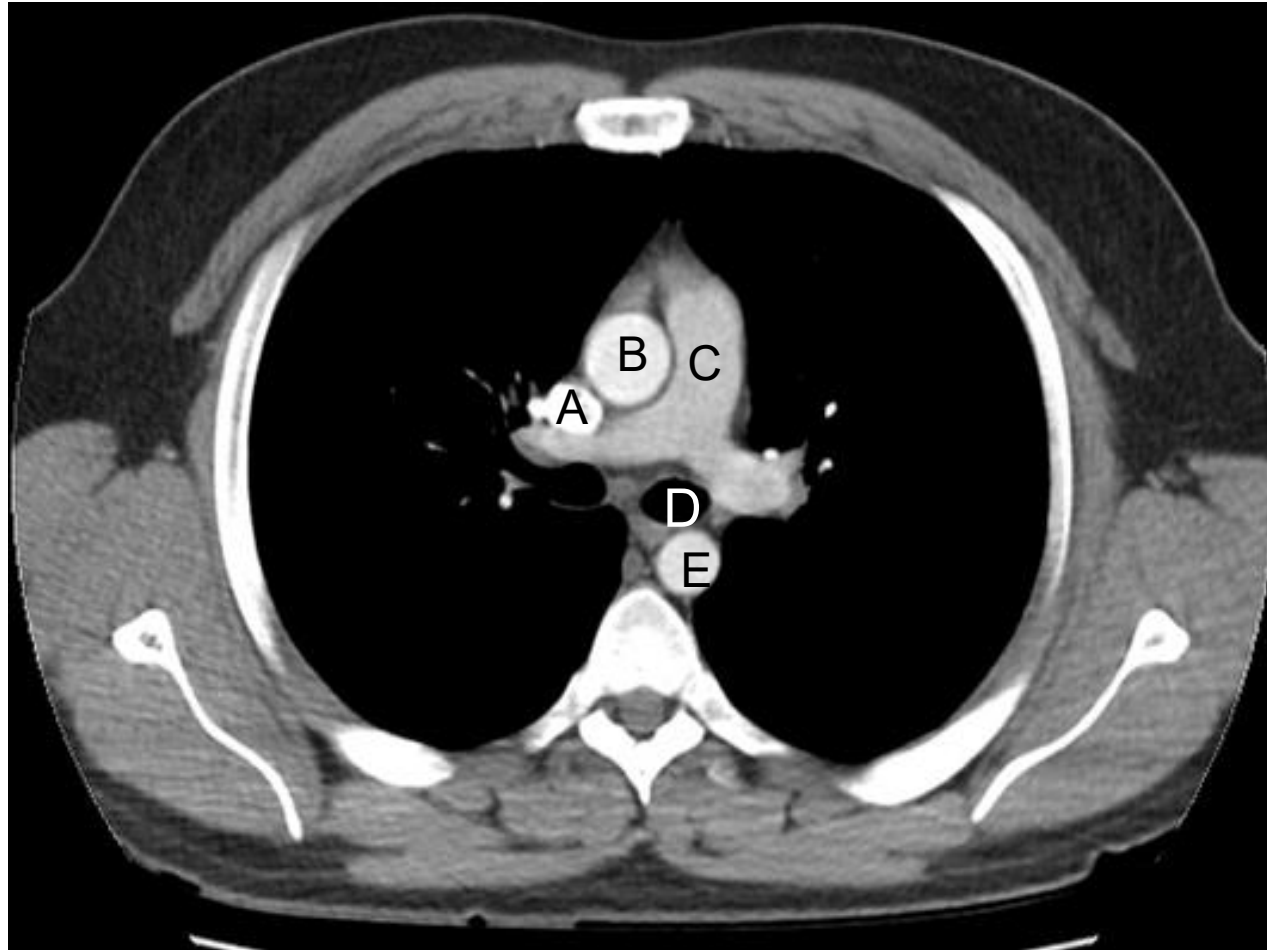
- A. Aortic Sac
- B. First arch artery
- C. 4<sup>th</sup> arch artery
- D. Dorsal aorta
- E. Truncus arteriosus



Dixon A, Normal CT chest. Case study, Radiopaedia.org (Accessed on 25 Mar 2024)  
<https://doi.org/10.53347/rID-36676>

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**4. A 58-year-old man comes to the emergency department because of a 2-hour history of palpitations and light-headedness. He has a history of hypertension and dyslipidemia. His pulse is 110/min and irregularly irregular, and blood pressure is 135/85 mm Hg. An ECG shows atrial fibrillation with rapid ventricular response. He is stabilized with medications, but his arrhythmia persists despite optimal medical therapy. A catheter-based radiofrequency ablation of the natural pacemaker of the heart with the placement of a permanent ventricular pacemaker is planned. Given the area that is being ablated, which of the following is the most likely embryological origin of this area?**

- A. Right horn of sinus venosus
- B. Left horn of sinus venosus
- C. Primitive atrium
- D. Primitive ventricle
- E. Bulbus cordis
- F. Truncus arteriosus
- G. Endocardial cushion

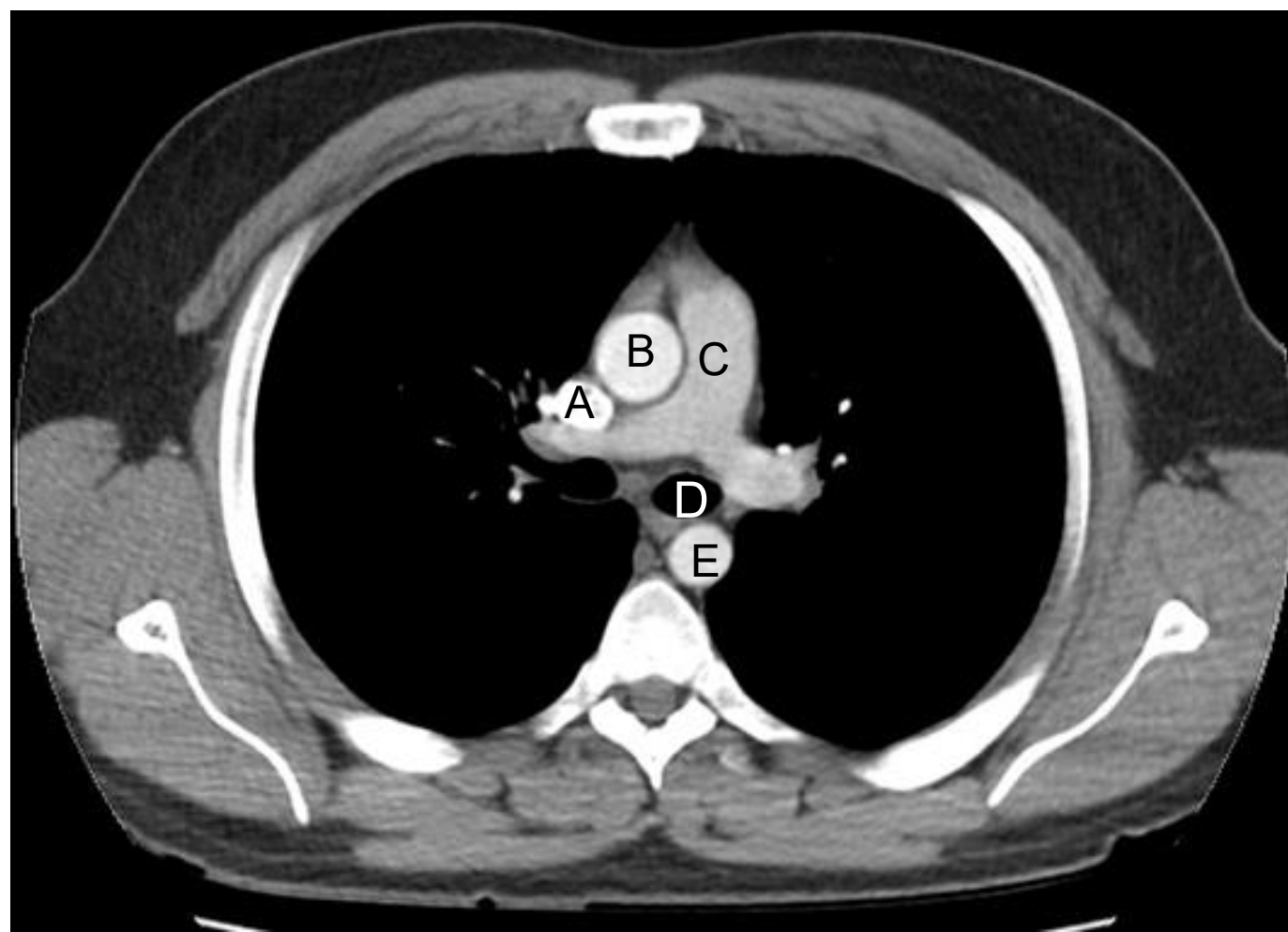
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- C. Primitive atrium
- D. Primitive ventricle
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- F. Truncus arteriosus
- G. Endocardial cushion



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- A. A
- B. B
- C. C
- D. D
- E. E

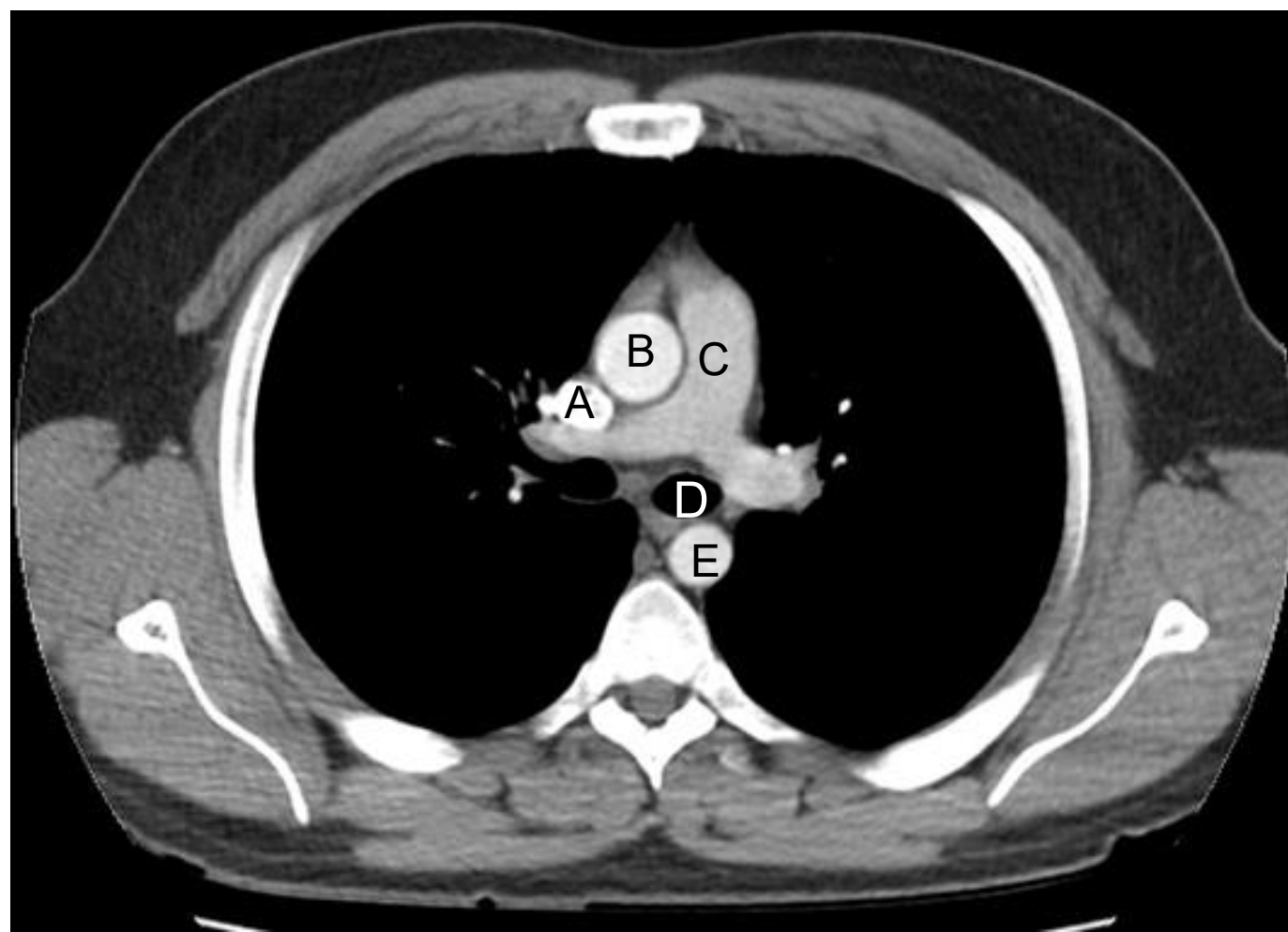


Dixon A, Normal CT chest. Case study, Radiopaedia.org (Accessed on 25 Mar 2024)  
<https://doi.org/10.53347/rID-36676>



5. A 72-year-old man is brought to the emergency department because of a 2-hour history of intense midback pain, nausea, sweating, and feelings of faintness. Blood pressure measured in the right arm is markedly elevated at 220/105 mm Hg. A transesophageal echocardiography shows the presence of an aortic dissection flap within the descending aorta, while the ascending aorta is unaffected. A CT scan of a normal chest is shown. Which of the following labeled structures is the most likely location of the initial tear that led to the findings in this patient?

- A. A
- B. B
- C. C
- D. D
- E. **E**



Dixon A, Normal CT chest. Case study, Radiopaedia.org (Accessed on 25 Mar 2024)  
<https://doi.org/10.53347/rID-36676>